

Revised Generation Interconnection System Impact Study Report

For Queue Project Q17-0890

Rosita Creek 138 kV
September 18, 2018

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q17-0890. The Interconnection Customer proposes a 406.3 MW solar photovoltaic facility interconnecting at the Rosita Creek 138 kV substation in the New Jersey area. The request was received on November 09, 2017. The study evaluated system intact and single contingency conditions using the regional planning case, including 409 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 24 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$56,755,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Glenmore / Blanchardville 138 kV line: 147.0% loading (290 MVA) following the outage of the Coronado / Mojave Siphon 345 kV; DFAX = 48.2%.

The Blanchardville / Gateway 138 kV is overloaded to 144.1% of its 304 MVA emergency rating for the loss of the Mojave Siphon / Coronado Substation 345 kV. The project contributes 48.2% DFAX (196.0 MW) to this overload.

The Gateway / Rosita Crk 138 kV is overloaded to 141.7% of its 318 MVA emergency rating for the loss of the Mojave Siphon / Coronado 345 kV. The project contributes 48.2% DFAX (196.0 MW) to this overload.

Walnut-Engle Tap / Smoky Hill 138 kV: 135.5% loading (243 MVA) following the outage of the Mojave Siphon 345 kV/138 kV XFMR; DFAX = 8.5%.

The Glenmore / Plaza Walmart 138 kV is overloaded to 126.9% of its 211 MVA emergency rating for the loss of the Hawthorne / Russell Energy Center 138 kV. The project contributes 33.1% DFAX (134.3 MW) to this overload.

For the loss of Walnut-Engle Tap / Cabinet Gorge 138 kV, the Roman Nose / Passons 138 kV L loads to 125.7% of its 248 MVA rating. Project DFAX: 8.9%.

Forest Hill / Smoky Hill 138 kV line: 124.8% loading (313 MVA) following the outage of the Mojave Siphon 345-138 kV Transformer Bank; DFAX = 8.5%.

For the loss of Walnut-Engle Tap / Cabinet Gorge 138 kV, the Delmont / Passons Substation 138 kV loads to 123.8% of its 244 MVA rating. Project DFAX: 8.9%.

Brownfield City / Forest Hill 138 kV L: 122.9% loading (329 MVA) following the outage of the Mojave Siphon 345 kV/138 kV Auto; DFAX = 8.5%.

Delmont / El Camino Tap 138 kV Line: 120.0% loading (241 MVA) following the outage of the Walnut-Engle Tap / Cabinet Gorge 138 kV; DFAX = 8.9%.

For the loss of Russell Energy Ctr / Hawthorne Substation 138 kV, the Brownfield City Substation / Plaza Walmart 138 kV loads to 119.1% of its 216 MVA rating. Project DFAX: 33.1%.

El Camino Tap / Purcell 138 kV Line: 118.0% loading (235 MVA) following the outage of the Walnut-Engle Tap / Cabinet Gorge 138 kV; DFAX = 8.9%.

For the loss of Gateway / Rosita Crk 138 kV, the Rosita Creek / Grandview 138 kV Line loads to 113.3% of its 341 MVA rating. Project DFAX: 100.0%.

For the loss of Russell Energy Ctr / Fair Lawn Substation 138 kV line, the W Pt Municipal Substation / Cabinet Gorge 138 kV L loads to 112.7% of its 312 MVA rating. Project DFAX: 11.5%.

Manor / Black Hollow 138 kV Line: 112.0% loading (276 MVA) following the outage of the West Pt Municipal / Yahara Riv 138 kV; DFAX = 6.9%.

The Hawthorne / Purcell 138 kV L is overloaded to 110.9% of its 215 MVA emergency rating for the loss of the Walnut-Egle Tap / Cabinetgorge 138 kV. The project contributes 8.9% DFAX (36.3 MW) to this overload.

The Fair Lawn / Manor 138 kV line is overloaded to 109.9% of its 303 MVA emergency rating for the loss of the W Pt Municipal / Yahara Riv 138 kV. The project contributes 6.9% DFAX (27.9 MW) to this overload.

For the loss of Walnut-Engle Tap / Cabinet Gorge 138 kV, the Hawthorne / Russell Energy Center Substation 138 kV loads to 108.5% of its 293 MVA rating. Project DFAX: 8.9%.

The Roman Nose / Glenmore 138 kV is overloaded to 107.7% of its 215 MVA emergency rating for the loss of the Glenmore / Plaza Walmart 138 kV. The project contributes 35.7% DFAX (144.9 MW) to this overload.

For the loss of W Pt Municipal / Yhara Riv 138 kV, the Fair Lawn / Russell Energy Center Substation 138 kV loads to 105.0% of its 385 MVA rating. Project DFAX: 6.9%.

For the loss of Russell Energy Ctr / Fair Lawn 138 kV, the West Point Municipal / Yahara Riv Substation 138 kV loads to 103.9% of its 553 MVA rating. Project DFAX: 8.1%.

For the loss of Gateway / Rosita Creek 138 kV, the Vesper / Gamebird 138 kV loads to 103.8% of its 206 MVA rating. Project DFAX: 8.9%.

The Coronado / Grandview 138 kV line is overloaded to 102.2% of its 361 MVA emergency rating for the loss of the Gateway / Rosita Crk 138 kV. The project contributes 100.0% DFAX (406.3 MW) to this overload.

The Tocaloma Junction / Vesper 138 kV line is overloaded to 100.0% of its 185 MVA emergency rating for the loss of the Gateway / Rosita Creek 138 kV Line. The project contributes 8.9% DFAX (36.0 MW) to this overload.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Glenmore / Blanchardville 138 kV	Rebuild line with higher-capacity conductor	\$6,402,000	88% (\$5,630,000)
Gateway / Blanchardville 138 kV Line	Rebuild line with higher-capacity conductor	\$5,688,000	88% (\$5,002,000)
Gateway / Rosita Crk 138 kV	Rebuild line with higher-capacity conductor	\$6,183,000	88% (\$5,437,000)
Walnut-Engle Tap / Smoky Hill 138 kV	Rebuild line with higher-capacity conductor	\$1,600,000	66% (\$1,055,000)
Glenmore / Plaza Walmart 138 kV	Rebuild line with higher-capacity conductor	\$5,214,000	78% (\$4,066,000)
Roman Nose / Passons 138 kV L	Rebuild line with higher-capacity conductor	\$4,206,000	55% (\$2,313,000)
Forest Hill / Smoky Hill 138 kV line	Rebuild line with higher-capacity conductor	\$2,673,000	66% (\$1,762,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Passons / Delmont 138 kV	Rebuild line with higher-capacity conductor	\$2,476,000	55% (\$1,362,000)
Forest Hill / Brownfield City 138 kV	Rebuild line with higher-capacity conductor	\$8,148,000	66% (\$5,371,000)
El Camino Tap / Delmont 138 kV	Rebuild line with higher-capacity conductor	\$3,317,000	55% (\$1,824,000)
Brownfield City Substation / Plaza Walmart 138 kV	Rebuild line with higher-capacity conductor	\$3,081,000	78% (\$2,403,000)
El Camino Tap / Purcell 138 kV Line	Rebuild line with higher-capacity conductor	\$3,730,000	55% (\$2,051,000)
Rosita Crk Substation / Grandview Substation 138 kV line	Rebuild line with higher-capacity conductor	\$7,457,000	100% (\$7,457,000)
Cabinet Gorge / W Point Municipal 138 kV	Rebuild line with higher-capacity conductor	\$2,584,000	49% (\$1,259,000)
Manor / Black Hollow 138 kV Line	Rebuild line with higher-capacity conductor	\$4,188,000	44% (\$1,864,000)
Hawthorne / Purcell 138 kV L	Rebuild line with higher-capacity conductor	\$2,321,000	55% (\$1,276,000)
Fair Lawn / Manor 138 kV line	Reconductor line	\$1,375,000	44% (\$612,000)
Russell Energy Ctr / Hawthorne Substation 138 kV	Reconductor line	\$482,000	31% (\$149,000)
Roman Nose / Glenmore 138 kV	Reconductor line	\$3,956,000	95% (\$3,765,000)
Russell Energy Center / Fair Lawn 138 kV L	Reconductor line	\$556,000	44% (\$247,000)
West Point Municipal / Yahara River 138 kV L	Reconductor line	\$602,000	50% (\$303,000)
Veper / Gamebird 138 kV	Reconductor line	\$1,094,000	46% (\$504,000)
Grandview / Coronado 138 kV line	Reconductor line	\$628,000	100% (\$628,000)
Tocaloma Junction / Vesper 138 kV line	Reconductor line	\$902,000	46% (\$415,000)
	Total allocated to project		\$56,755,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.